

WELSH GOVERNMENT

Modelling Team

Model Accuracy Assessment Report

main.py (v1) — Random Forest Regressor on Welsh Health Service Data

Dual-Audience Report | Technical & Executive Versions

May 2026

Data and Model Status

Data: Trimmed organisational dataset — not the full available series. Results are not generalisable beyond the data provided.

Model performance is materially below acceptable thresholds for policy use (see Section 4). This report does not endorse use of current model outputs for operational or commissioning decisions.

SECTION A — EXECUTIVE VERSION

1. Executive Summary

This report presents the results of an accuracy assessment of a Random Forest regression model (main.py, version 1) trained on Welsh health service data. The purpose of the assessment was to determine whether the model produces forecasts of sufficient quality to be used in analysis.

The short answer is: the model does not currently meet an acceptable accuracy standard.

Headline Results

MAE: 151,662.26 — on average the model's predictions are wrong by around 152,000 units

RMSE: 217,659.53 — large errors are common; the model struggles with high-value observations

MAPE: 291.81% — predictions deviate from actuals by almost three times the actual value on average

R²: 0.2263 — the model explains only 22.6% of the variation in the data

These figures indicate that the model is performing poorly. A MAPE of 291.81% means that, on average, the model's prediction for any given month is nearly three times higher or lower than the true value. An R² of 0.2263 means the model accounts for less than a quarter of what drives the numbers it is trying to predict.

Why the Model Is Underperforming

The most likely cause is that the training data has been trimmed. The dataset covers six different health service areas with very different value ranges — from counts in the tens to figures exceeding 800,000. A model trained on a truncated version of this data will not have seen enough of the range of outcomes to generalise reliably.

A secondary contributor is that the model is being asked to forecast across multiple, heterogeneous series simultaneously. Values on very different scales — for example a percentage target (between 0 and 100) alongside raw patient counts (potentially in the hundreds of thousands) — create a difficult prediction problem that drives up the MAPE in particular.

What Happens Next

- The full, untrimmed dataset should be used for retraining.
- Separate models should be considered for each data series or series type.
- The model must not be used to inform policy or operational decisions until these issues are addressed and accuracy re-assessed.

ASSUMPTION: 'Trimmed' data is taken to mean that the dataset provided for training is a subset of a larger available series, either in time coverage or in the number of series included.

2. Background and Purpose

The Welsh Government modelling team developed main.py (v1) to forecast values across a combined dataset of Welsh health service indicators. The dataset brings together six organisational data sources covering diagnostics waiting times, emergency department attendances, pathway of care delays, cancer treatment timelines, and referral-to-treatment waiting lengths.

The purpose of this accuracy assessment is to determine whether the model produces forecasts reliable enough to support analytical or policy work. This report documents the assessment and its conclusions.

Data Sources in Scope

diagnostic_pathway_waiting_times.csv	(252 rows, Jan 2019 – Dec 2025)
ed_attendances.csv	(84 rows, Jan 2019 – Dec 2025)
ed_attendances_targets.csv	(252 rows, Jan 2019 – Dec 2025)
pathway_of_care_delays.csv	(66 rows, Apr 2023 – Dec 2025)
patients_starting_cancer_treatment_within_62_days.csv	(299 rows, Jun 2019 – Dec 2025)
rtt_patient_waiting_length.csv	(420 rows, Jan 2019 – Dec 2025)
Total records in training dataset: 1,373	

3. Methodology Summary

The model uses a Random Forest — an approach that makes predictions by running data through many decision trees and averaging the results. It was trained on historical monthly data from the sources above, with the most recent 20% of records held back for testing.

To help the model learn from patterns over time, each data point was enriched with features representing what had happened in previous months: values from 1, 2, 3, 6, and 12 months prior, plus rolling averages over 3 and 6 months. The model was trained on the earliest 80% of data and tested on the remaining 20%, in time order, so that it was always predicting into the future rather than the past.

ASSUMPTION

The 80/20 chronological split means the test set covers approximately the most recent 15 months of data (roughly late 2024 – Dec 2025). The exact test period depends on the sorted distribution of records across all series.

4. Key Findings

All figures below are taken directly from Results.txt. No values have been adjusted or rounded beyond what appears in the source file.

Metric	Value	Interpretation	Assessment
MAE (Mean Absolute Error)	151,662.26	~152k units average error	Moderate concern
RMSE	217,659.53	Penalises large errors heavily	High concern
MAPE	291.81%	Extreme relative error	Very high concern
R ² (R-squared)	0.2263	Explains only 22.6% of variance	Insufficient

What the Metrics Mean in Plain English

The MAE of 151,662 tells us that for a typical prediction, the model is wrong by around 152,000 units. For a series where values run into the hundreds of thousands, this may seem proportionate, but the MAPE of 291.81% reveals the real problem: on series where values are small — percentages, small counts — the model's errors are enormous relative to the actual values.

The R² of 0.2263 is the most decisive indicator. A value of 1.0 would mean perfect prediction; 0.0 would mean the model is no better than simply predicting the average every time. At 0.2263, the model is only marginally better than a naïve average baseline.

Feature Importance

The most influential predictors were rolling averages and lag features — particularly the 6-month rolling mean (importance: 0.132) and the 3-month rolling mean (0.118). This confirms that recent historical values carry most of the signal, which is expected for health service time series. The model is learning from recency, but not well enough to produce accurate forecasts.

5. Limitations and Caveats

Primary Limitation — Trimmed Data

The training dataset has been trimmed. This is the most significant constraint on model performance.

A model trained on an incomplete series cannot reliably generalise to the full range of values it will encounter.

Until the full dataset is used, accuracy results should be treated as a lower-bound estimate of what the model might achieve, not as a stable baseline.

- The dataset combines series with fundamentally different scales and units: raw patient counts, delay counts, and percentage targets are mixed. This makes a single joint model hard to optimise.
- `pathway_of_care_delays.csv` covers only April 2023 to December 2025 — far shorter than other series. Lag features requiring 12 months of history will be unavailable for much of this series.
- All data is organisational rather than official statistics and has not been quality-assured to National Statistics standards.
- Geography is constant (Wales only) and sex and age are both 'all persons / all ages' throughout. These features add no discriminating power and inflate the feature space without benefit.
- The MAPE of 291.81% is heavily influenced by series with very small values (e.g. percentage metrics close to zero). This metric alone should not be used to condemn the model; it should be assessed per-series.

6. Recommendations

Immediate Actions

- Do not use current model outputs to inform operational or commissioning decisions.
- Provide the full, untrimmed dataset for retraining and re-run the accuracy assessment before any further use.

Model Improvements

- Consider training separate models per data series or per source file, rather than a single joint model across heterogeneous series.
- Evaluate whether percentage-based series (e.g. ED target compliance) should be modelled separately or transformed before inclusion.
- Assess model performance per series (per-series MAE, MAPE, R^2) rather than only reporting aggregate metrics, to identify which series the model handles well and which it does not.
- Consider a naïve seasonal benchmark (same month, prior year) as a minimum comparator. At $R^2 = 0.2263$, the current model may not outperform it.

Reporting

- Any future use of model outputs must be accompanied by per-series confidence intervals and clearly caveated as based on organisational rather than official data.

SECTION B — TECHNICAL VERSION

1. Executive Summary (Technical)

main.py (v1) implements a Random Forest Regressor (sklearn, n_estimators=300, max_features='sqrt', random_state=42) trained on a combined organisational dataset of 1,373 monthly Welsh health service records spanning January 2019 to December 2025. An 80/20 chronological train-test split was applied. The model was evaluated against a held-out test set using MAE, RMSE, MAPE, and R^2 .

Results Summary

MAE: 151,662.26

RMSE: 217,659.53

MAPE: 291.81%

R^2 : 0.2263

Assessment: model performance is materially insufficient for analytical or policy use in its current form.

The primary driver of poor performance is the use of a trimmed dataset. Secondary drivers include the heterogeneity of series value ranges, the short coverage of one source file (pathway_of_care_delays.csv, 66 rows from April 2023), and the inclusion of uninformative one-hot encoded features (geography, sex, age are constant across all records).

2. Background and Purpose (Technical)

The assessment was conducted to evaluate whether main.py (v1) produces forecasts of sufficient accuracy for use in Welsh health service analysis. The model is a supervised regression model using engineered time series features; it is not a native time series model (e.g. ARIMA or Prophet) and does not enforce stationarity or model seasonality explicitly.

Dataset Characteristics

Total records: 1,373
Date range: January 2019 – December 2025
Geography: Wales (constant — single value throughout)
Sex: Persons (constant)
Age: All ages (constant)
Source files: 6
Unique descriptions: 14
Value range: 51.9 – 802,268 (mean: 105,451; median: ~24,971)
Value std dev: 178,674 (high variance across series types)

ASSUMPTION: The dataset labelled training_dataset.csv is a trimmed version of a larger available series. The model has not been trained on the full data, which is the primary cause of poor generalisation.

3. Methodology Summary (Technical)

Feature Engineering

Features were constructed per group (description + geography + sex + age). Given that geography, sex, and age are constant throughout, grouping is effectively by description only — equivalent to per-series feature construction.

- Lag features: lag_1, lag_2, lag_3, lag_6, lag_12 (value shifted n months within group)
- Rolling mean features: rolling_3, rolling_6 (shift(1).rolling(n).mean() to prevent leakage)
- Date parts: year, month
- One-hot encoded: description, geography, sex, age, source_file

Data Leakage Check

The code applies shift(1) before rolling means, correctly preventing target leakage into rolling features.

Chronological sort is applied before the 80/20 split, correctly preserving temporal order.

No shuffling is applied. The approach is sound for preventing future leakage.

Model Configuration

- Algorithm: RandomForestRegressor (scikit-learn)
- n_estimators: 300
- max_features: 'sqrt' (number of features considered at each split)
- random_state: 42 (reproducible)
- n_jobs: -1 (parallelised)
- No hyperparameter tuning documented — defaults used for max_depth, min_samples_split, etc.

Train / Test Split

An 80/20 chronological split was applied after sorting all records by date. The training set comprises the first 80% of records by date; the test set the remaining 20%. Given 1,373 records post-feature-engineering (actual post-dropna count will be lower due to lag initialisation), the test set is approximately 275 records.

ASSUMPTION

The exact post-dropna record count is not reported in Results.txt. The figures above are approximations based on the raw CSV row count (1,373). Lag and rolling features will have removed early rows from each series, reducing the usable count.

Evaluation Metrics

- MAE: Mean Absolute Error — average absolute deviation between prediction and actual.
- RMSE: Root Mean Squared Error — penalises large errors more than MAE; sensitive to outliers.

- MAPE: Mean Absolute Percentage Error — relative error; inflated when actuals are near zero.
- R^2 : Coefficient of determination — proportion of variance explained by the model.

4. Key Findings (Technical)

All figures are reproduced exactly from Results.txt.

Metric	Value	Interpretation	Assessment
MAE (Mean Absolute Error)	151,662.26	~152k units average error	Moderate concern
RMSE	217,659.53	Penalises large errors heavily	High concern
MAPE	291.81%	Extreme relative error	Very high concern
R ² (R-squared)	0.2263	Explains only 22.6% of variance	Insufficient

Metric Interpretation

MAE: 151,662.26

The average absolute prediction error is 151,662 units. Given the dataset mean of ~105,451 and median of ~24,971, this MAE exceeds the median value by a factor of approximately 6. The error magnitude is dominated by high-value series (e.g. `rtt_patient_waiting_length.csv` records reaching 802,268).

RMSE: 217,659.53

RMSE of 217,660 is substantially higher than MAE (151,662), indicating the presence of large outlier errors. The ratio $RMSE/MAE \approx 1.43$ suggests a fat-tailed error distribution — the model performs poorly on a meaningful subset of predictions, not just on average.

MAPE: 291.81%

A MAPE of 291.81% is extreme and likely driven by series where actual values are small (e.g. percentage-based metrics close to zero or low-count series). When the denominator in MAPE is small, even a modestly wrong prediction produces a very large percentage error. This metric should be disaggregated by series before drawing conclusions about specific data types.

R²: 0.2263

An R² of 0.2263 is the most important summary metric. It indicates that the model explains approximately 22.6% of the variance in the test set. For context, a naïve mean predictor yields $R^2 = 0.0$; a seasonal naïve baseline (predict same month, prior year) would likely produce $R^2 > 0.2263$ on this dataset type. This should be tested explicitly.

Feature Importances

The top 10 features by Gini importance are:

Rank	Feature	Importance Score
1	rolling_6 (6-month rolling mean)	0.1318
2	rolling_3 (3-month rolling mean)	0.1180
3	lag_3 (value 3 months prior)	0.1105
4	lag_12 (value 12 months prior)	0.1034
5	lag_6 (value 6 months prior)	0.1015
6	description: patients waiting <26 weeks (all stages of pathway)	0.0930
7	lag_2 (value 2 months prior)	0.0852
8	lag_1 (value 1 month prior)	0.0847
9	source_file: rtt_patient_waiting_length.csv	0.0745
10	year	0.0439

Rolling and lag features dominate, together accounting for approximately 73.5% of total importance. This confirms the model is learning from recent history rather than from categorical descriptors. The high rank of rolling_6 (0.132) and rolling_3 (0.118) over lag_1 (0.085) suggests medium-term trend is more predictive than the immediate prior month in this data.

The presence of source_file_rtt_patient_waiting_length.csv (importance 0.075) as a top-10 feature indicates the RTT series has a distinct enough distribution to materially alter predictions — consistent with it being the largest and highest-value series in the dataset (420 rows, values up to 802,268).

5. Limitations and Caveats (Technical)

Primary Limitation — Trimmed Training Data

The dataset is explicitly noted as trimmed. A Random Forest trained on a truncated series will have an incomplete view of the outcome distribution, leading to:

- Underfitting on unseen value ranges
- Biased lag and rolling features (early values in each series are artefacts of truncation, not true history)
- Unreliable feature importance rankings

This limitation cannot be mitigated without access to the full dataset.

Additional Technical Limitations

- Heterogeneous target scale: values range from 51.9 to 802,268 with a standard deviation of 178,674. A single Random Forest must learn decision boundaries across radically different value scales simultaneously. Tree-based models do not normalise inputs, so this is handled implicitly via tree depth and splits, but with reduced efficiency.
- MAPE inflation: the MAPE of 291.81% is not directly comparable to a model trained on a single series. Percentage-based series (e.g. '% of ED attendances spending less than 4 hours') will have actual values in the range 0–100 and will dominate the MAPE calculation when predicted values deviate even modestly.
- pathway_of_care_delays.csv coverage: 66 rows from April 2023 only. Lag_12 features will be unavailable for the first 12 months of this series; lag_6 for the first 6. After dropna, only records from April 2024 onwards would contribute to this series, leaving approximately 21 usable rows before the train/test split.
- Constant categorical features: geography ('Wales'), sex ('persons'), and age ('all ages') are constant throughout. One-hot encoding these adds columns with zero discriminating power and increases the feature space without benefit. These should be dropped before retraining.
- No hyperparameter optimisation: the model uses sklearn defaults for max_depth (unlimited), min_samples_split (2), and min_samples_leaf (1). Unlimited depth increases the risk of overfitting to training series.
- No cross-validation: a single 80/20 split was used. Walk-forward cross-validation would give a more robust estimate of out-of-sample performance.
- No per-series evaluation: aggregate metrics mask series-specific performance. A series where the model performs well is indistinguishable in aggregate from one where it does not.
- Data is organisational, not official statistics. No quality assurance to National Statistics standards has been applied.

6. Recommendations (Technical)

Immediate

- Retrain on the full, untrimmed dataset. This is the highest-priority action. Current results are not a valid baseline until this is done.
- Run per-series evaluation (MAE, RMSE, MAPE, R^2 per description and per source_file) to identify which series the model handles well and which it does not.
- Implement a naïve seasonal benchmark (same month, prior year) and compare R^2 against it. At 0.2263, the current model may not outperform it.

Model Architecture

- Remove constant categorical features (geography, sex, age) from the feature set before retraining.
- Consider stratified or separate models per source file or per description group, particularly to isolate percentage-based series from count-based series.
- Add walk-forward cross-validation (minimum 3 folds) to replace the single 80/20 split for a more robust performance estimate.
- Tune max_depth and min_samples_leaf to control overfitting; use out-of-bag error as a cheap internal estimate.

Evaluation

- Add per-series MAPE to the evaluation output to understand which series drive the aggregate MAPE of 291.81%.
- Report prediction intervals alongside point forecasts (e.g. using quantile regression forests or conformal prediction wrappers).
- Document the post-dropna training set size explicitly in the results output so the effective sample size is traceable.

7. Appendix

A. Raw Results (from Results.txt)

Reproduced verbatim from Results.txt

MAE: 151662.26

RMSE: 217659.53

MAPE: 291.81%

R²: 0.2263

Top 10 feature importances:

rolling_6	0.131813	
rolling_3	0.117988	
lag_3	0.110516	
lag_12	0.103388	
lag_6	0.101524	
description_patients waiting less than 26 weeks (all stages of pathway)	0.092985	
lag_2	0.085183	
lag_1	0.084742	
source_file_rtt_patient_waiting_length.csv	0.074543	
year	0.043922	

B. Dataset Summary (derived from training_dataset.csv)

Training Dataset Profile

Total rows: 1,373

Columns: value, description, date, geography, sex, age, source_file

Date range: 2019-01-01 to 2025-12-01

Geography: Wales (only)

Sex: Persons (only)

Age: All ages (only)

Unique descriptions: 14

Source files: 6

Value min: 51.9

Value max: 802,268

Value mean: 105,451

Value std dev: 178,674

C. Acceptance Check Record

Check	Status	Note
All figures match inputs provided	PASS	Exact figures from Results.txt used throughout
Limitations not watered down	PASS	Section 5 is explicit and unmodified
Executive summary stands alone	PASS	Section A.1 is self-contained

Check	Status	Note
No invented numbers or findings	PASS	All numbers sourced from Results.txt / CSV inspection
Assumptions explicitly labelled	PASS	3 assumptions flagged in body
Data trimming limitation carried through	PASS	Primary limitation in Sections 5A and 5B

D. Glossary

- MAE: Mean Absolute Error — average of the absolute differences between predictions and actual values.
- RMSE: Root Mean Squared Error — square root of the average squared differences; penalises large errors more than MAE.
- MAPE: Mean Absolute Percentage Error — average absolute error as a percentage of the actual value; sensitive to small actual values.
- R^2 : Coefficient of determination — proportion of variance in the target explained by the model. Range: $-\infty$ to 1; 1 = perfect, 0 = naïve mean baseline.
- Random Forest: an ensemble of decision trees whose predictions are averaged; robust to overfitting relative to a single tree.
- Lag feature: the value of a time series at n periods prior to the current observation.
- Rolling mean: the average of the previous n observations; here computed after shifting by 1 to prevent leakage.
- Walk-forward validation: cross-validation that respects temporal ordering by training only on past data and testing on future data.
- Gini importance: a measure of how much each feature reduces impurity across all splits in the forest.